

AWS RDS SQL Server Health Check — Key Steps for Optimal Performance

When performing a **health check on AWS RDS SQL Server**, the goal is to ensure database **availability, performance, security, and cost optimization**. Below are the structured steps:

1 Check RDS Instance Status

- Verify that the RDS instance is in the **“Available”** state (via AWS Console or CLI).
- Check **uptime, maintenance window, and failover history**.
- Review **instance class (CPU/RAM)** to ensure it matches workload demand.

2 Review CloudWatch Metrics

- Monitor key performance indicators:
 - **CPU Utilization**
 - **Freeable Memory**
 - **Read/Write Latency**
 - **Disk Queue Depth**
 - **Database Connections**
- Set or validate **CloudWatch alarms** for threshold breaches.

3 Check Storage & IOPS Utilization

- Verify **allocated storage vs used space**.
- Monitor **IOPS consumption** and **burst balance (for GP2/GP3 storage)**.
- Ensure **storage auto-scaling** is enabled if workload growth is expected.

4 Validate SQL Server Error Logs

- Connect via SSMS → RDS instance → Management → SQL Server Logs.
- Review for:
 - Failed logins
 - Deadlocks or critical errors
 - Repeated restarts or failed backups

5 Database Status & Integrity

- Run queries to ensure all databases are:
 - Online and accessible
 - Not in **“Restoring”** or **“Suspect”** state
- Validate **DBCC CHECKDB** integrity reports via SQL Agent or manual execution.

6 Backup & Restore Validation

- Check **RDS automated backups** are enabled and running successfully.
- Validate **retention period** (default 7 days — can extend to 35).
- Test **manual snapshot** creation and restore process.
- Confirm **transaction log backups** (for full recovery mode) are functioning.

7 Performance Tuning Checks

- Review **top resource-consuming queries** using:
 - AWS Performance Insights
 - sys.dm_exec_query_stats
- Analyze **wait stats**, **blocking sessions**, and **missing indexes**.
- Rebuild/reorganize indexes where fragmentation > 30%.
- Update outdated statistics.

8 Parameter Group & Configuration Review

- Validate key RDS parameters:
 - max server memory
 - max degree of parallelism (MAXDOP)
 - cost threshold for parallelism
 - remote admin connections
- Ensure parameter group is applied and consistent across replicas.

9 High Availability & DR Verification

- If using **Multi-AZ**, verify:
 - Standby instance health and replication lag (minimal or zero)
 - Recent failover test results
- For **Read Replicas**, confirm replication status and latency via RDS console or AWS CLI.

10 Security & Access Controls

- Validate:
 - RDS **security group** inbound rules (restricted to approved IPs/VPCs)
 - **IAM policies** for least-privilege access
 - **Encryption** (at rest using KMS, in transit via SSL/TLS)
 - No public accessibility unless required
 - Strong password policy for RDS master user

1 1 Maintenance & Patching

- Check **RDS maintenance window** schedule.
- Confirm **automatic minor version upgrades** are enabled (if policy allows).
- Validate recent patch or engine version applied.

1 2 Monitoring & Alerting

- Ensure **Enhanced Monitoring** is enabled for OS-level metrics.
- Integrate with **AWS CloudWatch Alarms**, **EventBridge**, or third-party monitoring (e.g., SolarWinds, Datadog).
- Confirm email/SNS notifications are functioning.

1 3 Cost & Performance Optimization

- Review:
 - Instance type utilization (upgrade/downgrade recommendations)
 - Storage type and throughput vs performance requirements

- Reserved instance or savings plan opportunities
- Unused read replicas or snapshots

Summary Checklist

Area	Key Focus
Availability	Instance state, Multi-AZ health
Performance	CPU, memory, latency, query tuning
Security	IAM, encryption, access control
Backups	Automated backups, snapshots
Monitoring	CloudWatch, Enhanced Monitoring
Configuration	Parameter groups, MAXDOP, memory
Cost	Sizing, optimization, reserved usage